Intensive Care Medicine Experimental

Unbiased miRNA biomarker discovery for Detection of Severe Sepsis

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Full Title:	Unbiased miRNA biomarker discovery for Detection of Severe Sepsis
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Abstract:	Background: Sepsis is a life-threatening condition with high morbidity and mortality. Early diagnosis of sepsis remains challenging due to the limited specificity of conventional biomarkers. Circulating microRNAs (miRNAs) have emerged as promising candidates for sepsis detection and monitoring due to their regulatory roles in inflammation and immune response. Methods: We conducted a prospective observational study at a quaternary academic medical center in Southeast Asia, enrolling 116 patients with neurological hemorrhage admitted to the Intensive Care Unit or High-Dependency Unit. Patients were stratified into non-sepsis, sepsis, and systemic inflammatory response syndrome (SIRS) without microbiological confirmation groups. Serum samples were collected on Days 1, 3, and 5 of fever spike. High-throughput miRNA profiling was performed using the ID3EAL™ PanoramiR panel, followed by normalization and differential expression analysis. Candidate miRNAs were identified using statistical filtering, and pathway enrichment analysis was conducted to explore biological relevance. Results: Global circulating miRNA levels were comparable between sepsis and non-sepsis groups. However, distinct time-dependent expression patterns were observed. On Day 1, early sepsis was characterized by broad miRNA dysregulation, with miR-192-5p significantly upregulated and several miRNAs (including miR-485-5p and miR-493-5p) downregulated. By Day 3, miR-520a-3p emerged as the only significantly dysregulated miRNA, exhibiting downregulation. On Day 5, miR-98-5p was consistently downregulated, suggesting involvement in later-stage immune dysregulation. Pathway analysis implicated key inflammatory and metabolic pathways, including NF-κB, PI3K–AKT–mTOR signaling, and transcriptional regulation. Notably, miR-744-5p and let-7 family members demonstrated opposing expression patterns between early- and late-onset sepsis, suggesting stage-specific immune polarization. Conclusion: This study identifies dynamic, time-dependent miRNA signatures associated with sepsis progression. miR-192-5p, miR-520a-3p, and miR-98-5p represent potential stage-specific biomarkers for early detection and disease monitoring. These findings support the development of a temporally stratified miRNA panel to improve diagnostic precision and provide insights into sepsis pathophysiology.

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Unbiased miRNA biomarker discovery for Detection of Severe Sepsis

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Abstract

Background:

Sepsis is a life-threatening condition with high morbidity and mortality. Early diagnosis of sepsis remains challenging due to the limited specificity of conventional biomarkers. Circulating microRNAs (miRNAs) have emerged as promising candidates for sepsis detection and monitoring due to their regulatory roles in inflammation and immune response.

Methods:

We conducted a prospective observational study at a quaternary academic medical center in Southeast Asia, enrolling 116 patients with neurological hemorrhage admitted to the Intensive Care Unit or High-Dependency Unit. Patients were stratified into non-sepsis, sepsis, and systemic inflammatory response syndrome (SIRS) without microbiological confirmation groups. Serum samples were collected on Days 1, 3, and 5 of fever spike. High-throughput miRNA profiling was performed using the ID3EAL™ PanoramiR panel, followed by normalization and differential expression analysis. Candidate miRNAs were identified using statistical filtering, and pathway enrichment analysis was conducted to explore biological relevance.

Results:

Global circulating miRNA levels were comparable between sepsis and non-sepsis groups. However, distinct time-dependent expression patterns were observed. On Day 1, early sepsis was characterized by broad miRNA dysregulation, with miR-192-5p significantly upregulated and several miRNAs (including miR-485-5p and miR-493-5p) downregulated. By Day 3, miR-520a-3p emerged as the only significantly dysregulated miRNA, exhibiting downregulation. On Day 5, miR-98-5p was consistently downregulated, suggesting involvement in later-stage immune dysregulation. Pathway analysis implicated key inflammatory and metabolic pathways, including NF-κB, PI3K–AKT–mTOR signaling, and transcriptional regulation. Notably, miR-744-5p and

let-7 family members demonstrated opposing expression patterns between early- and late-onset sepsis, suggesting stage-specific immune polarization.

Conclusion:

This study identifies dynamic, time-dependent miRNA signatures associated with sepsis progression. miR-192-5p, miR-520a-3p, and miR-98-5p represent potential stage-specific biomarkers for early detection and disease monitoring. These findings support the development of a temporally stratified miRNA panel to improve diagnostic precision and provide insights into sepsis pathophysiology.

Introduction

Sepsis is a life-threatening condition resulting from a dysregulated immune response to infection, causing systemic inflammation and organ failure. It is a significant cause of morbidity and mortality in ICUs. Early detection of sepsis can greatly improve patient outcomes, but current biomarkers, like C-reactive protein (CRP) and procalcitonin (PCT), often lack specificity, particularly in critically ill patients with underlying conditions.

MicroRNAs (miRNAs) are small, non-coding RNA molecules that play crucial roles in post-transcriptional regulation of gene expression, influencing a wide range of biological processes including immune function, inflammation, and cell apoptosis [1]. These molecules, approximately 22 nucleotides in length, bind to messenger RNAs (mRNAs) and inhibit their translation or promote degradation, thereby modulating protein production [2]. Over recent years, there has been increasing evidence of miRNAs' involvement in various pathologies, including sepsis, a life-threatening systemic inflammatory response to infection with significant mortality and morbidity worldwide [3].

In the context of sepsis, miRNAs have emerged as both regulators and indicators of disease progression, offering insights into the pathophysiological mechanisms underlying this complex condition [4]. Several miRNAs are differentially expressed in septic patients and have been found to influence critical aspects of the immune response, inflammation, and tissue injury [5]. For instance, miRNAs such as miR-146a, miR-155, and miR-223 are key players in modulating inflammatory pathways and immune cell function during sepsis [6]. Additionally, dysregulated miRNAs in sepsis have been linked to adverse outcomes, including organ failure and increased mortality, positioning these molecules as potential biomarkers for early diagnosis and prognosis.

Furthermore, therapeutic modulation of specific miRNAs offers promising new avenues for managing sepsis. By targeting miRNAs that contribute to hyperinflammation or immune suppression, researchers aim to develop novel treatments that can mitigate the severity of sepsis or reduce the risk of complications [7]. Therefore, miRNAs present a compelling research frontier with potential applications in diagnosing, prognosticating, and treating sepsis, making them a focal point for translational studies in critical care.

In this study, we aim to identify specific miRNAs that are strongly associated with sepsis, based on the day of sepsis onset.

Materials & Methods

Study cohort

This prospective observational study was approved by the National Healthcare Group (NHG) Domain Specific Review Board (DSRB) (2020/00020) and conducted in accordance with ethical standards for human research. This study was conducted at the National University Hospital, Singapore (NUHS), from 25th November 2020 to 17th September 2024, recruiting 116 patients admitted to the ICU or HDU with brain hemorrhage. Eligible patients are over 21 years old with clinically or radiologically suspected or confirmed neurological hemorrhage, expected to stay for more than 48 hours. Exclusion criteria include hematological malignancy, neutropenia, immunocompromised state, pregnancy, and other factors that might affect the study outcome. Informed consent is obtained from a legally authorized representative (LAR) of each participant, ensuring time for decision-making. The diagnosis of sepsis is obtained from the treating physician or the commencement of antibiotics and confirmed by microbiological cultures that were routinely performed during this period as the standard of care to detect sepsis. The attending physician was involved in data analysis of immuno-parameters and white blood cell (WBC) infection score. Results are stratified based on patients who have no systemic inflammatory response syndrome (SIRS), SIRS but no microbiological evidence of sepsis (via blood, sputum, urine cultures, etc.), sepsis (SIRS + microbiological evidence) and retrospectively analyzed. Patients were treated according to the treating physician's routine standard of care practices in the best interest of the patient.

Clinical data collection

Comprehensive clinical data are collected and recorded in our electronic medical records system, covering demographic information, medical history, and the timing and presence of SIRS, sepsis, and septic shock. Disease severity is assessed using APACHE II and SOFA scores. Routine blood tests, blood culture timing, and therapeutic interventions (inotropes, antibiotics) are documented. Longitudinal data from Day 1 to Day 5 of temperature spike include parameters such as prothrombin time/activated partial thromboplastin time (PT/aPTT), full blood count (FBC), WBC infection scores, Intensive Care Infection Score (ICIS), and immune markers. Any complications, ICU/HDU length of stay, and mortality are also recorded. Figure 1a provides a visual outline of the study workflow, from patient enrollment to data collection, sample processing, and miRNA analysis.

Figure 1. a) Study workflow b) Serum sample collection timeline for sepsis and non-sepsis groups, with samples collected on Days 1, 3, and 5 of fever spike to facilitate comparative biomarker analysis across time points. c) Workflow for miRNA expression profiling using qPCR, illustrating sequential stages of miRNA isolation, cDNA conversion, pre-amplification, qPCR, and data analysis. This process includes quality control with RNA spike-in controls, 12 cycles of pre-amplification for enhanced detection, and high-throughput profiling in a 384-well assay panel, generating amplification curves for analysis of miRNA expression levels.

Blood collection, serum processing and storage

Over a period of 2 years, blood samples were collected from patients with neurological hemorrhage on three different days—Day 1, Day 3, and Day 5 of fever spike, from two distinct patient groups: those with sepsis (n=32) and those without sepsis (n=28). Each group follows the same schedule,

with samples taken at the same intervals to enable comparative analysis of biomarkers in septic and non-septic conditions over time (Figure 1b). Serum was extracted from these samples, aliquoted, and stored at -80°C, prior to subjecting them to batch laboratory tests. The remaining blood samples are stored at -80ºC over a period of 5 years for future biomedical research use.

Total RNA isolation and miRNA profiling

High-throughput miRNA profiling was conducted using the ID3EAL™ PanoramiR miRNA Knowledge Panel (Mirxes), which includes 376 miRNAs selected from miRBase 22, HMDD 3.2, and The Cancer Genome Atlas. Each assay has undergone extensive wet-lab validation using synthetic miRNA templates and RNA from biological samples. Briefly, 200 µL of frozen serum was thawed and subjected to RNA extraction using the Maxwell RSC miRNA Tissue Kit and the Maxwell RSC system (Promega) in accordance with the manufacturer's instructions. A proprietary ~20 nt spike-in control RNA (Mirxes), with sequences distinct from known human miRNAs (miRBase version 22), was added to the sample lysis buffer prior to RNA isolation. From the resulting 60 µL RNA eluate, 5 µL was used for each RT multiplex pool (A, B, C, or D) and reverse-transcribed to cDNA using conformation-restricted miRNA-specific RT primers and the ID3EAL™ cDNA Synthesis System (Mirxes) at the following RT parameters: 25°C for 10 min, 30°C for 10 min, 35°C for 10 min, 40°C for 10 min, followed by 95°C for 5 min. The resultant cDNAs were then divided into eight multiplex pools (A1, B1, C1, D1, A2, B2, C2, D2) and pre-amplified using the ID3EAL™ miRNA Augmentation System (Mirxes) under the following conditions: 25°C for 10 sec, 95°C for 10 min, 40°C for 5 min, followed by 12 cycles of 95°C for 10 sec and 60°C for 30 sec. qPCR master mix was added to cDNAs from each sample and loaded into the ID3EAL™ PanoramiR microRNA Knowledge Panel (Mirxes) prior to miRNA expression profiling using the QuantStudio5 Real-Time PCR system (Applied Biosystems). Upon the

completion of profiling, raw threshold cycle (Ct) values were determined using the QuantStudio Design and Analysis software with automatic baseline and threshold settings. Technical variations introduced during RNA isolation and the process of RT-qPCR were normalized using the spike-in control RNA. The Ct values of miRNAs from each sample are further normalized to the global miRNA expression. Log-2 transformed fold change in miRNA expression values of samples from cell culture media were calculated based on △△Ct values. The RT-qPCR-based miRNA expression profiling workflow is summarized in Figure 1c.

Statistical analyses

All statistical analyses were performed using R version 4.2.2. Raw Ct values were first normalized to spike-in RNA's Ct values, before global normalization. Log2 Foldchanges were calculated using normalized Ct, and statistical significance of findings (p-value) was analyzed by two-tailed unpaired Student's t-test.

To elucidate the biological functions and molecular pathways associated with the differentially expressed (DE) miRNAs, target identification was performed using R/Bioconductor's multiMiR package. Predicted targets were identified using 8 predictive databases: DIANA_microT, ElMMo, MicroCosm, miRanda, miRDB, PicTar, PITA and TargetScan. Validated targets were identified using 3 validated databases: miRecords, miRTarBase and TarBase. Target genes identified were used for pathway analysis using R/Bioconductor's enrichplot, ReactomePA, and clusterProfiler packages. Plots were generated using R's ggplot2.

Results

qPCR Ct Threshold-Based miRNA Detection in Sepsis vs. Non-Sepsis Cohorts

The qPCR threshold cycle (Ct) value, which is inversely proportional to miRNA expression, is used as a surrogate readout for circulatory miRNA abundance measured in each serum sample across different time points in both sepsis and non-sepsis patient groups (Figure 2a). To minimize false positives in high-throughput screens, we included miRNAs with Ct values from 9 to 33 for further analysis and excluded miRNAs with Ct values that exceed the set range. As shown in Figure 2b, the total circulatory miRNA counts detected from sepsis and non-sepsis samples were not statistically different, indicating that sepsis does not grossly alter circulatory miRNA biogenesis, secretion or turnover. Our data involving sepsis patients of the Asian population contradicts an earlier study that reported markedly lower levels of circulatory blood miRNAs in sepsis patients of the Western population[8] .

Figure 2. Total number of miRNAs expressed in serum samples from sepsis and non-sepsis patients. a) Scatter plot of global Ct values for various miRNAs across sepsis and non-sepsis groups on Days 1, 3, and 5. b) Total number of miRNAs detected by the PanoramiR Knowledge panels in samples from the non-sepsis and sepsis groups.

miRNA expression profile of patients with or without sepsis stratified by day

To determine whether miRNAs could serve as robust biomarkers that predicts early versus late onset sepsis, we stratified sepsis patients by the disease onset (day of fever spike) and analyzed their serum miRNA expression profiles. As shown in Supplementary Figure 1, miRNAs such as hsa-miR-744-5p, members of the let-7 family (let-7g-5p, let-7a-5p, let-7d-5p), hsa-miR-30e-5p, hsa-miR-379-5p and hsa-miR-107 appeared to be downregulated (Blue blocks in the heatmap) in

late-onset sepsis patients (fever spike at day 5) compared to early-onset sepsis patients (fever spike at day 1 or day 3). hsa-miR-744-5p and members of the let-7 family have been described as anti-inflammatory regulators. An important question, therefore, is why these microRNAs are downregulated in late-onset sepsis yet upregulated in early-onset sepsis. One possible explanation relates to differences in immune polarization across disease stages. Early-onset sepsis may be characterised by a predominantly Type 1 (Th1-biased) inflammatory response that is necessary for effective pathogen clearance, with upregulation of anti-inflammatory microRNAs serving as a regulatory counterbalance. In contrast, late-onset sepsis may reflect profound immune dysregulation, in which an exaggerated or maladaptive Type 2 (Th2-skewed) response emerges. Such Th2 dominance has been associated with immune paralysis, poorer clinical outcomes, and increased susceptibility to secondary infections, potentially explaining the observed downregulation of these anti-inflammatory microRNAs [9]. Conversely, members of the hsa-miR-320 family (hsa-miR-320b, hsa-miR-320d) and other miRNAs (hsa-miR-642a-5p/345-5p/340-5p/30a-5p/19a-5p/187-3p/1246/628-5p/542-3p/582-5p/197-3p/1290/148a-3p/770-5p/129-5p/1291)exhibit increased expression (Red blocks in the heatmap) in late-onset sepsis patients (Supplementary Figure 1). Notably, miR-744-5p and several members of the let-7 family (let-7g-5p, let-7a-5p, let-7d-5p) exhibited consistent upregulation in early-onset sepsis patients, corroborating with their known pro-inflammatory roles in specific tissues and conditions by modulating key regulatory proteins in innate immune signaling pathways [10–12] acute inflammatory response. The panel initially included 376 miRNAs. However, 112 miRNAs were excluded as they were expressed in less than 5% of the cohort (n=59), leaving 264 miRNAs for further analysis.

Early-onset sepsis associated miRNAs (Day 1)

Among these, 240 miRNAs showed no dysregulation. Of the 24 dysregulated miRNAs, 9 were excluded due to a lack of statistical significance ($p > 0.05$). Consequently, 15 candidate miRNAs were selected for further analysis (Figure 3).

Figure 3. Flowchart outlining the selection process of candidate miRNAs using the ID3EAL qPCR PanoramiR Assay Panel for Day 1.

On Day 1, several miRNAs show significant downregulation in sepsis, including miR-485-5p ($p = 0.00110$), miR-134-5p, miR-493-5p ($p = 0.0106$), miR-432-5p ($p = 0.0133$), miR-411-5p ($p = 0.0202$), miR-369-3p ($p = 0.0312$), and miR-487b-3p ($p = 0.0458$), all exhibiting negative log2 fold changes. Conversely, miR-192-5p is significantly upregulated in sepsis on Day 1, with a positive log2 fold change ($p = 0.0331$). Supplementary Figure 2 illustrates these differentially expressed miRNAs in a volcano plot, while Tables 1A and 1B list the top 10 upregulated and downregulated miRNAs, respectively. Consistent with these findings, the heatmap in Supplementary Figure 3 highlights significant upregulated miRNAs in the sepsis group, such as hsa-miR-192-5p, hsa-miR-92b-3p, and hsa-miR-208b-3p, and downregulated miRNAs, including hsa-miR-485-5p, hsa-miR-487b-3p, and hsa-miR-493-5p.

Figures 4 and 5 show the pathway analysis for the dysregulated miRNAs on day 1 of sepsis. Gene targets of the upregulated miRNAs are shown in Figure 4a. The pathways regulated by these genes are shown in Figure 4b. Similarly, Figure 5a shows the identified gene targets of the downregulated miRNAs while the pathway analysis of these genes is shown on Figure 8b. Notably, the miRNAs downregulated in the Sepsis group on Day 1 post-fever spike are found to regulate several pathways including neuronal development pathways, wound healing and cell growth pathways.

Table 1A: Top 10 upregulated miRNAs in cells		
miRNA ID	**Log_2 Fold Change**	**p-value**
hsa-miR-206	1.51	0.107505
hsa-miR-193b-3p	1.24	0.097851
hsa-miR-208b-3p	1.11	0.220699
hsa-miR-92b-3p	1.11	0.065856
hsa-miR-192-5p	1.01	0.033104

Table 1B: Top 10 downregulated miRNAs in cells		
miRNA ID	**Log_2 Fold Change**	**p-value**
hsa-miR-379-5p	-1.5	0.013147
hsa-miR-487b-3p	-1.38	0.045845
hsa-miR-432-5p	-1.35	0.013331

hsa-miR-493-5p	-1.27	0.010628
hsa-miR-369-3p	-1.25	0.03118
hsa-miR-411-5p	-1.21	0.020244
hsa-miR-410-3p	-1.19	0.016048
hsa-miR-494-3p	-1.16	0.006169
hsa-miR-409-3p	-1.16	0.026138
hsa-miR-485-5p	-1.15	0.001096

Figure 4. a) miRNA-gene target network of miRNAs upregulated on day 1 post-fever spike in patients with Sepsis Yellow lines represent genes identified in validated databases, while cyan lines represent genes identified in predictive databases. b) Bubble plot of pathway analysis of the gene targets identified in (a)

Figure 5. a) miRNA-gene target network of miRNAs downregulated on day 1 post-fever spike in patients with Sepsis Yellow lines represent genes identified in validated databases, while cyan lines represent genes identified in predictive databases. b) Bubble plot of pathway analysis of the gene targets identified in (a)

Intermediate-onset sepsis associated miRNAs (Day 3)

From the remaining 264 miRNAs analyzed across 22 samples, 3 were found to be dysregulated (1 upregulated and 2 downregulated). After applying statistical criteria, 2 miRNAs were excluded for lack of significance ($p > 0.05$), leaving 1 candidate miRNA for further investigation (Figure 6).

Figure 6. Flowchart illustrating the selection process of a candidate miRNA using the ID3EAL qPCR PanoramiR Assay Panel on Day 3

As shown in Supplementary Figure 4 and Table 2A, miR-520a-3p ($p = 0.034356$) is the only significantly dysregulated miRNA identified in this analysis, exhibiting marked downregulation in the sepsis group on Day 3. This finding suggests a potential role for miR-520a-3p in the biological response to sepsis at this stage.

Table 2A: Top 10 downregulated miRNAs in cells		
miRNA ID	**Log_2 Fold Change**	**p-value**
hsa-miR-520a-3p	-1.23	0.034356
hsa-miR-190a-5p	-1.07	0.154494

These findings align with the heatmap results in Supplementary Figure 5. hsa-miR-154-5p shows significant upregulation in sepsis samples, highlighted by strong red coloring, indicating high expression in response to sepsis. Conversely, hsa-miR-520a-3p is downregulated in sepsis samples, marked by blue coloring, reflecting decreased expression on Day 3 under septic conditions. hsa-miR-190a-5p displays variable expression, with mixed upregulation and downregulation across samples, indicating a less consistent pattern in relation to sepsis.

Late-onset sepsis associated miRNAs (Day 5)

Among the remaining 264 miRNAs analyzed in a sample of 13, 14 were identified as dysregulated (5 upregulated and 9 downregulated). Following statistical evaluation, 13 miRNAs were excluded due to lack of significance ($p > 0.05$), leaving 1 candidate miRNA for further investigation (Figure 7).

Figure 7. Flowchart outlining the selection process of candidate miRNAs using the ID3EAL qPCR PanoramiR Assay Panel in Day 5

miR-98-5p is the only miRNA significantly dysregulated in the sepsis group on Day 5, showing notable downregulation ($p = 0.037382$) (Supplementary Figure 6 and Table 3A). This finding suggests that miR-98-5p may be involved in the response to sepsis on this specific day, potentially serving as a late-stage biomarker or regulatory factor in sepsis progression.

Table 3A: Top 10 downregulated miRNAs in cells

miRNA ID	Log_2 Fold Change	p-value
hsa-miR-551a	-1.7	0.255906
hsa-miR-98-5p	-1.68	0.037382
hsa-miR-508-3p	-1.64	0.208452
hsa-miR-592	-1.59	0.082002
hsa-miR-708-5p	-1.44	0.207418
hsa-miR-330-3p	-1.19	0.14251
hsa-miR-297	-1.15	0.3724
hsa-miR-23c	-1.05	0.317092
hsa-miR-1299	-1	0.60838

This is in congruent with findings from the heatmap (Supplementary Figure 7) which shows that hsa-miR-98-5p is significantly downregulated in sepsis samples, indicating it may be suppressed in the septic condition on Day 5.

GO analysis of upregulated miRNAs on day 5 post-fever spike in patients with sepsis revealed pathways including RNA processing, protein synthesis, and metabolic adaptation during the later stages of sepsis. This reflects a shift from inflammatory activation to cellular remodeling and repair (Figures 8 a and b).

GO analysis of miRNAs downregulated on day 5 of sepsis relieves repression of transcriptional, metabolic, and stress-response networks. This pattern supports a transition from acute inflammation to immune dysregulation and tissue injury, consistent with late-phase sepsis biology (Figures 9a and b).

Figure 8. a) miRNA-gene target network of miRNAs upregulated on day 5 post-fever spike in patients with Sepsis Yellow lines represent genes identified in validated databases, while cyan lines represent genes identified in predictive databases. b) Bubble plot of pathway analysis of the gene targets identified in (a)

Figure 9. a) miRNA-gene target network of miRNAs upregulated on day 5 post-fever spike in patients with Sepsis Yellow lines represent genes identified in validated databases, while cyan lines represent genes identified in predictive databases. b) Bubble plot of pathway analysis of the gene targets identified in (a)

Discussion

This study explores the differential expression of miRNAs in patients with and without sepsis across three clinically relevant timepoints (Day 1, 3, 5 of fever spike). Our findings highlight distinct miRNA expression patterns that may reflect specific molecular responses to sepsis with some miRNAs displaying temporal variability.

On Day 1, miR-192-5p was robustly upregulated, while multiple miRNAs including miR-485-5p, miR-493-5p, miR-432-5p, miR-411-5p, and miR-487b-3p were downregulated. This suggests that early sepsis is characterized by a broad reprogramming of miRNA networks, reflecting both activation of inflammatory cascades and suppression of regulatory pathways. Wang and Meng et

al found that miR-744-5p is significantly upregulated in sepsis, correlating with the activation of the NF-κB pathway, a central regulator of inflammation. This suggests miR-744-5p's involvement in amplifying the immune response during the hyper-inflammatory phase of sepsis [13]. miR-30e-5p has been associated with apoptosis, oxidative stress, and inflammation in a recent meta-analysis done to evaluate the potential of miRNAs as biomarkers for sepsis detection [14] GO analysis of day 1 upregulated miRNAs (Figure 8a and b) showed that they were significantly linked to diseases of signal transduction by growth-factor receptors and second messengers and RNA polymerase II-specific DNA-binding transcription-factor binding. miR-192-5p, miR-92b-3p, and miR-208-3p were linked to validated targets (RELA, STAT3[15], MAPK, PIK3R1) that are canonical members of mTOR[16, 17], NF-κB[18–20], and PI3K-Akt[17] pathways. NF-κB is known to be a regulator of pro-inflammatory gene transcription. Studies have found that pathogen-associated molecular patterns (PAMP) activate Toll-like receptors (TLRs) leading to the activation of NF-κB. This increases the transcription of cytokines like TNF-α, IL-1β, IL-6, coagulation factors and adhesion molecules, resulting in systemic inflammation and endothelial dysfunction[21], potentiating septic shock. NF-κB was found to be elevated in septic shock and is correlated with worse outcomes[21]. The PI3K/AKT pathway is crucial in regulating cell survival, growth, and immune cell function. mTOR, which is downstream of the PI3K/AKT pathway, regulates metabolism by integrating nutrient, energy, and stress signals. When dysregulated, it can result in impaired immunity and organ injury[22]. Interestingly, NF-κB regulated cytokines can stimulate AKT signaling; metabolic stress and mTOR dysregulation may feedback to influence NF-κB activation or inflammatory signaling.

By Day 3, miR-520a-3p showed consistent downregulation in sepsis. Interestingly, miR-520a-3p has previously been implicated in immune regulation and stress responses (Zhang et al. 2018),

raising the possibility that its downregulation contributes to ongoing immune dysregulation during the subacute phase of sepsis.

By Day 5, miR-98-5p, remained significantly dysregulated, exhibiting consistent downregulation in sepsis patients. This suggests that miR-98-5p suppression may be involved in the later phase of sepsis, potentially reflecting immune exhaustion or maladaptive responses as the disease progresses. Prior studies have implicated miR-98-5p in modulating inflammatory signaling and endothelial function [23, 24], further supporting its biological plausibility as a late-stage biomarker. miR-98-5p suppresses IL-1β induced inflammation and apoptosis via CASP3 and inhibits NF-κB signaling, thereby protecting against sepsis-related cardiac or lung injury[24, 25]. Conversely, upregulated miRNAs on day 5 like miR-92b-3p have been linked to increased mTOR signaling[16]. Studies have shown a PI3K–AKT–mTOR-linked metabolic reprogramming and immune-phase shifts in sepsis [22, 26], which may reflect adaptive efforts to restore homeostasis and transcriptional control.

The broad dysregulation in miRNAs on Day 1, with subsequent narrowing to specific miRNAs over time shows the dynamic and time-dependent nature of the host response in sepsis. Early dysregulation amplifies the inflammation cascades, while late suppression indicates immune exhaustion, endothelial injury, and post-septic tissue remodeling. This suggests that miRNAs may serve not only as static diagnostic markers but also as longitudinal indicators of disease progression and treatment response [4, 6] (The consistent involvement of specific miRNAs such as miR-192-5p (early), miR-520a-3p (intermediate), and miR-98-5p (late) provides a potential framework for developing a time-stratified biomarker panel that could improve the accuracy of sepsis detection and monitoring in critically ill patients.

In our cohort, miR-744-5p and multiple let-7 family members show an opposing expression pattern in early- versus late-onset sepsis (EOS vs LOS), raising the possibility that these miRNAs track stage-specific immune polarization rather than "sepsis" as a single static state. Early-onset sepsis is generally associated with a Type 1 (Th1-biased) inflammatory response, which is necessary for effective pathogen clearance[27, 28]. In this early phase, upregulation of anti-inflammatory microRNAs such as let-7 family members, which are known to suppress TLR and NF-κB signalling, may function as a negative feedback mechanism to limit excessive inflammation [29]. In contrast, late-onset sepsis is increasingly recognised as a state of immune dysregulation and immunosuppression, characterised by impaired Th1 responses and a relative shift toward Th2-skewed or anti-inflammatory pathways, which have been linked to poorer outcomes and increased susceptibility to secondary infections[4, 27]. The downregulation of miR-744-5p and let-7 family members in late-onset sepsis may therefore reflect this maladaptive immune state rather than resolution of inflammation.

Strengths and Limitations

A major strength of this study is the systematic, longitudinal collection of miRNA data across multiple days and conditions, providing a robust dataset that captures temporal changes in miRNA expression. This approach enhances our ability to identify candidate biomarkers that reflect dynamic disease processes rather than static snapshots. Additionally, the use of a high-throughput miRNA panel and rigorous statistical analysis strengthens the reliability of our findings.

However, several limitations warrant consideration. First, the cohort size, though informative, may limit the generalizability of our findings. A larger cohort would increase the statistical power and enable a more comprehensive assessment of miRNA expression patterns across a broader range of

clinical presentations. Moreover, while the study identifies several miRNAs with potential as biomarkers, further validation in external cohorts and experimental studies are necessary to confirm their functional roles in sepsis and SAH.

Conclusion

In conclusion, our study identifies a set of miRNAs that are differentially expressed in sepsis in a time-dependent manner, with miR-192-5p, miR-520a-3p, and miR-98-5p emerging as key candidates across distinct phases of illness. These findings support the utility of circulating miRNAs as promising biomarkers for the early detection and monitoring of sepsis and lay the groundwork for future studies exploring their functional significance and translational potential in critical care.

Declaration

Ethics approval and consent to participate

This study was approved by the National Healthcare Group (NHG) Domain Specific Review Board (DSRB) (2020/00020) and conducted in accordance with ethical standards for human research.

Consent for publication

Not applicable.

Availability of data and materials

The dataset(s) supporting the conclusions of this article is(are) included within the article (and its additional file(s)).

Competing interests

There are no competing interests among the authors.

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This study is not funded.

Author's contributions

CJLL was involved with the writing, editing and data analysis of this study. GWLE was involved with the data analysis, data curation and writing of this study. CLSY was involved with the data analysis and curation of this study. TSJ was involved with the writing, editing and data curation of this study. CJK was involved with the writing, editing, data curation and analysis of this study. NHWL was involved with the methodology, writing, editing, and conceptualization of this study.

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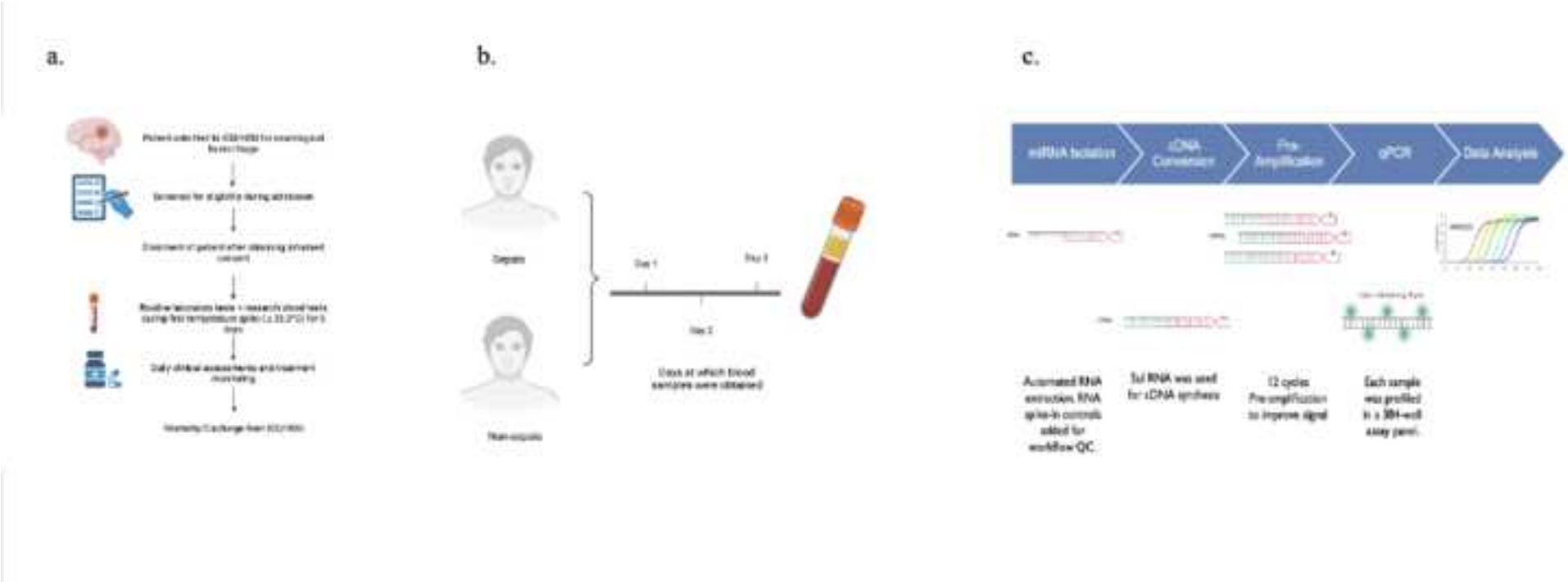

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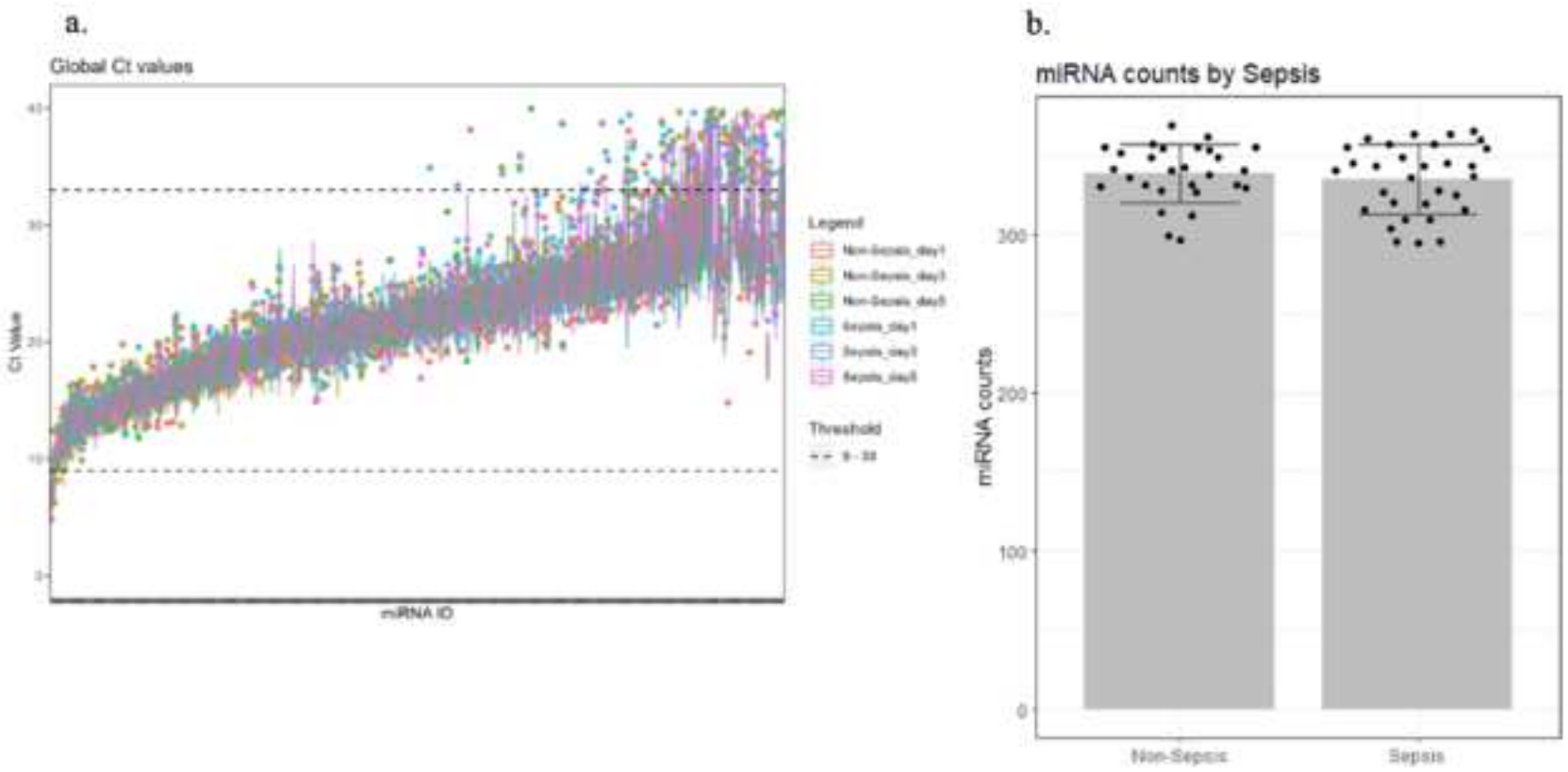

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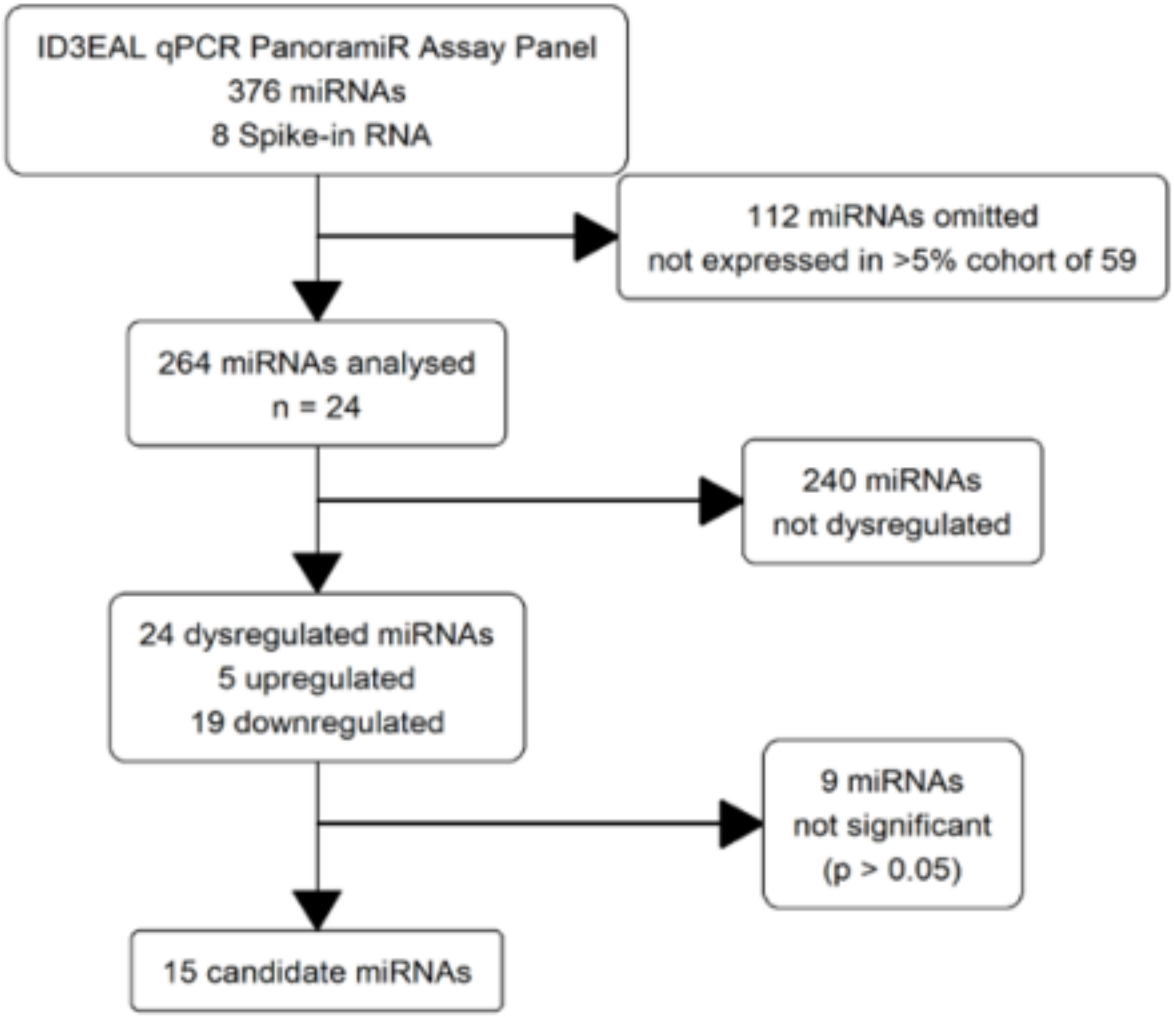

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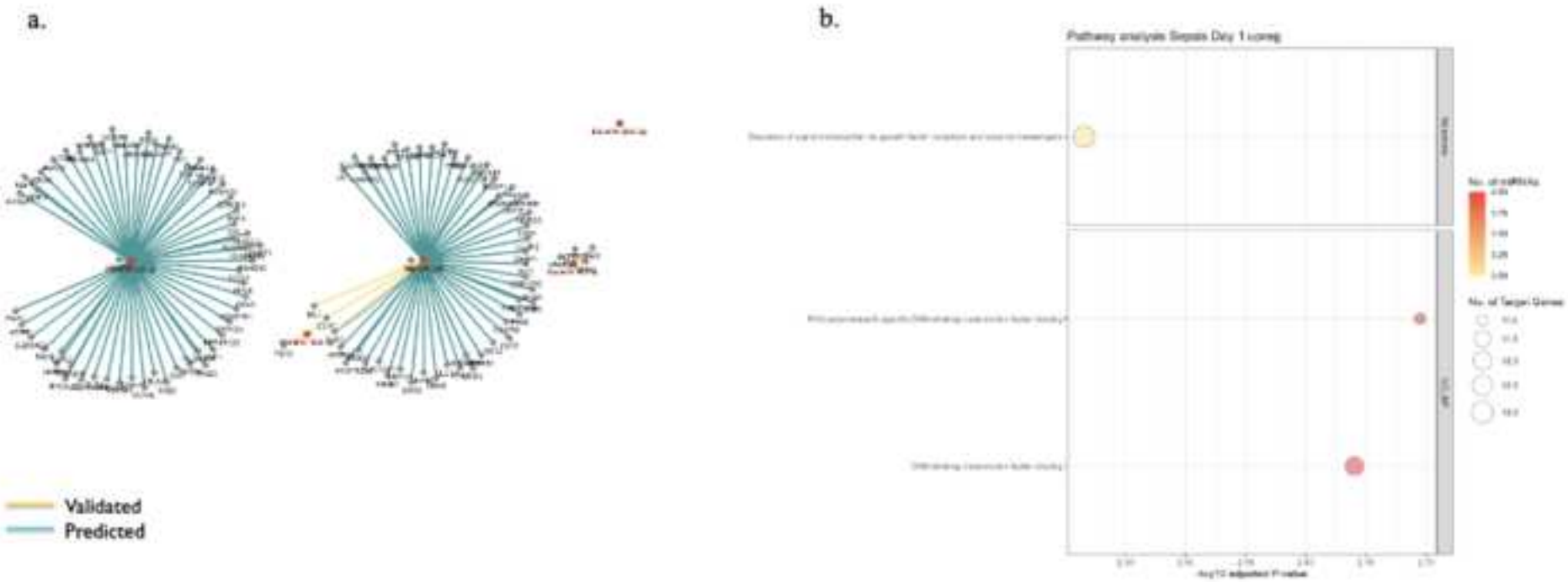

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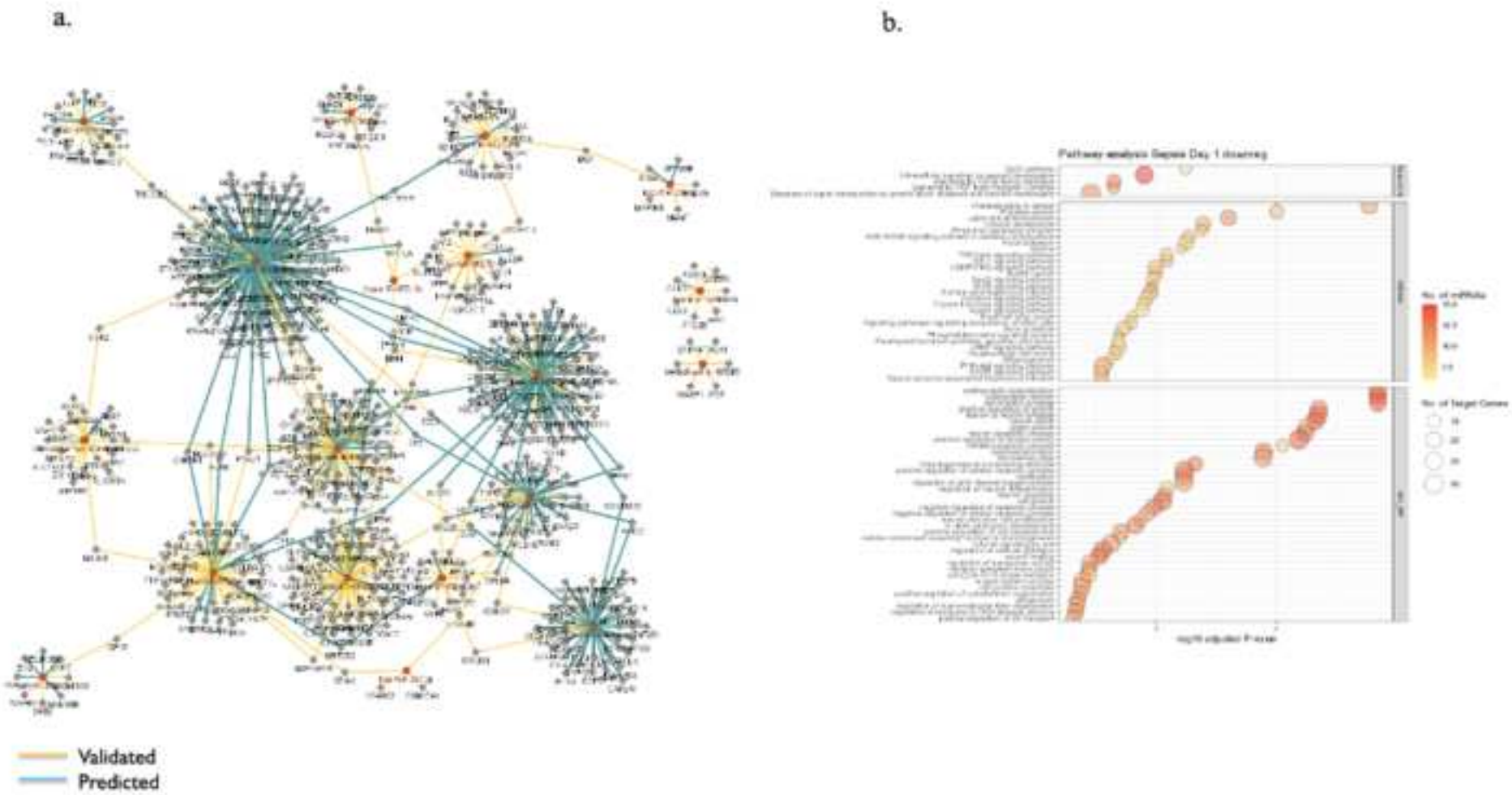

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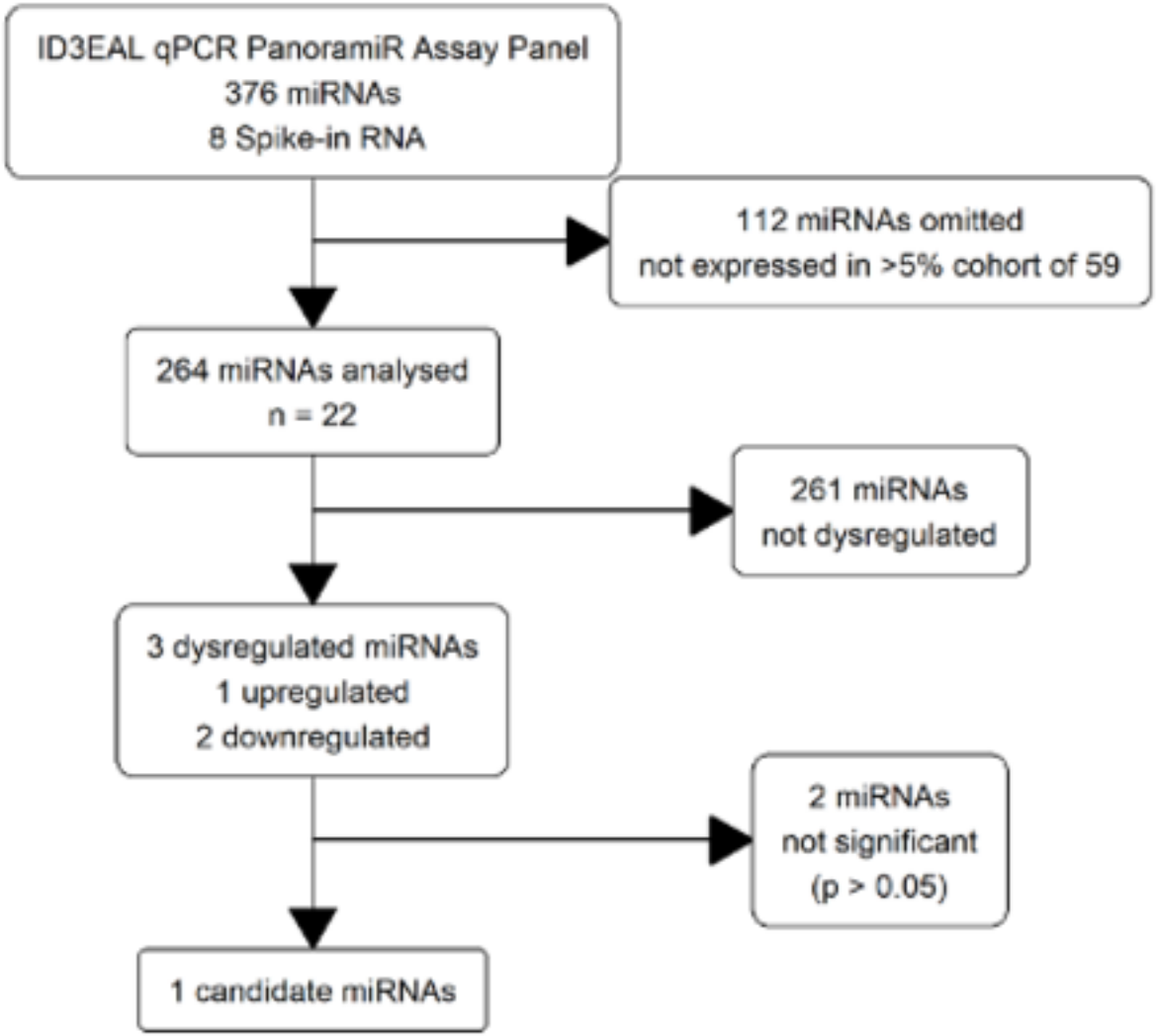

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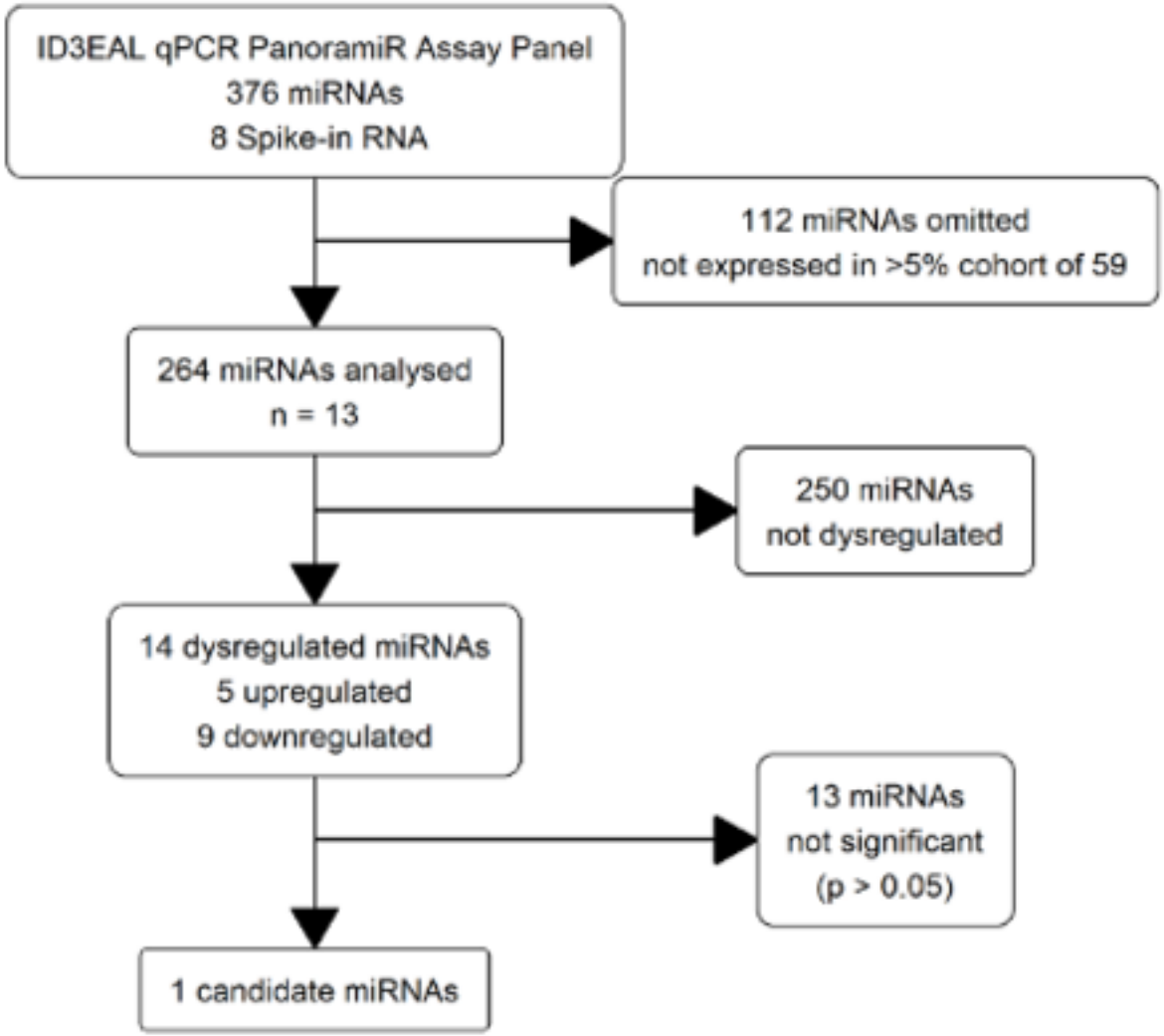

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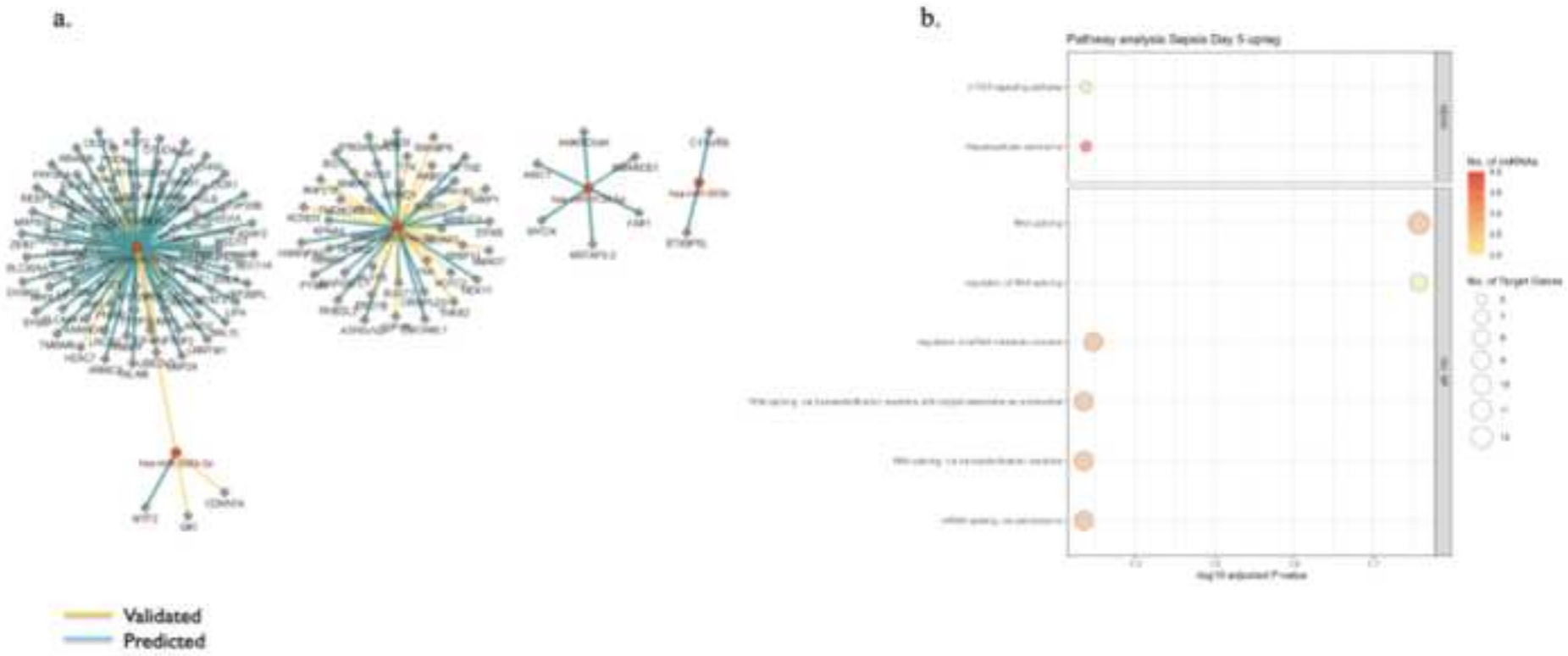

Figure

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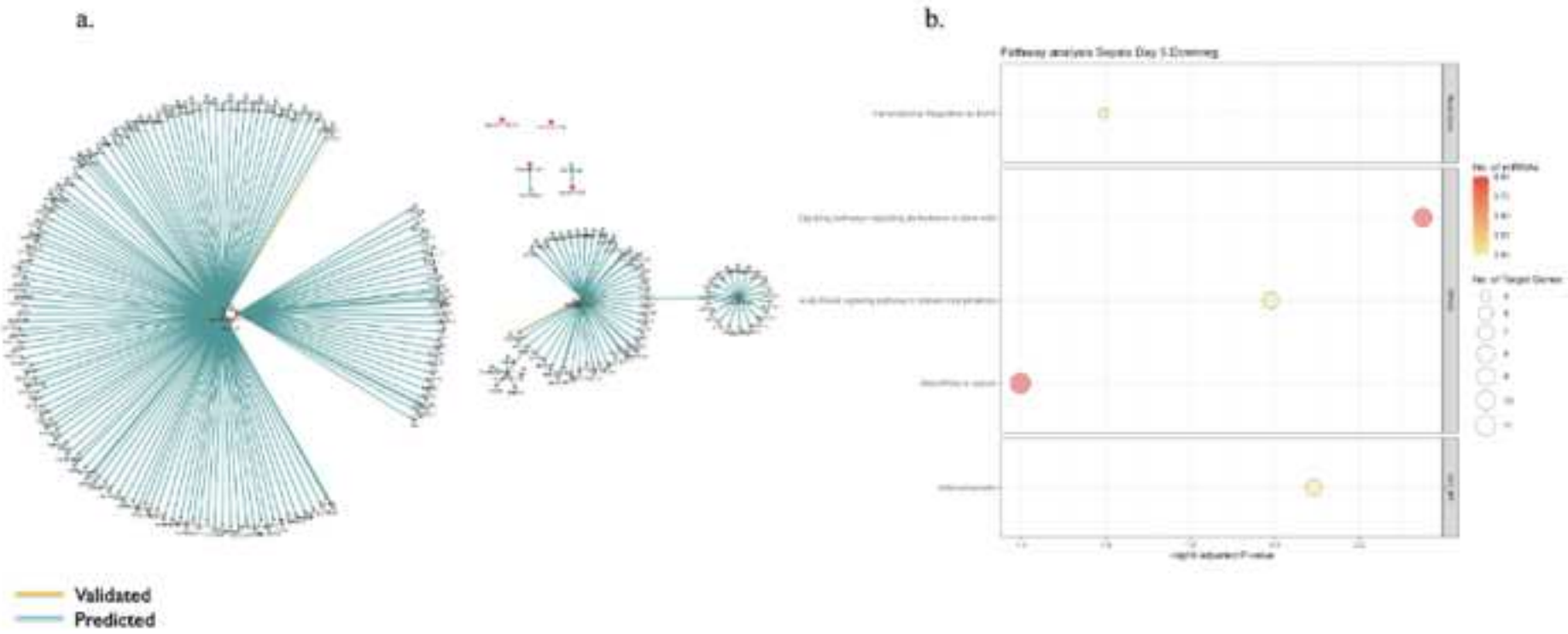

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